JEE 2009
Paper II

PART I : CHEMISTRY

SECTION – I

Single Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

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1. For a first order reaction A→P, the temperature (T) dependent rate constant (k) was found to follow the equation $\log k = -(2000)\frac{1}{T} + 6.0$. The pre-exponential factor A and the activation energy E_a, respectively, are

(A) 1.0×10^6 s^{-1} and 9.2 kJ mol^{-1}

(B) 6.0 s^{-1} and 16.6 kJ mol^{-1}

(C) 1.0×10^6 s^{-1} and 16.6 kJ mol^{-1}

(D) 1.0×10^6 s^{-1} and 38.3 kJ mol^{-1}

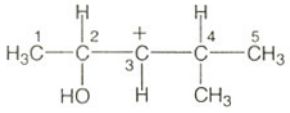
Answer ○ ○ ○ ● (A) (B) (C) (D)

2. The spin only magnetic moment value (in Bohr magneton units) of $Cr(CO)_6$ is

(A) 0 (B) 2.84 (C) 4.90 (D) 5.92

Answer ● ○ ○ ○ (A) (B) (C) (D)

3. In the following carbocation, H/CH_3 that is most likely to migrate to the positively charged carbon is

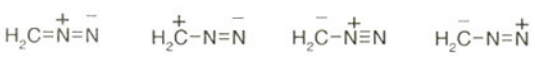


(A) CH_3 at C-4 (B) H at C-4

(C) CH_3 at C-2 (D) H at C-2

Answer ○ ○ ○ ● (A) (B) (C) (D)

4. The correct stability order of the following resonance structures is



(I) (II) (III) (IV)

(A) (I) > (II) > (IV) > (III) (B) (I) > (III) > (II) > (IV)

(C) (II) > (I) > (III) > (IV) (D) (III) > (I) > (IV) > (II)

Answer ○ ● ○ ○ (A) (B) (C) (D)

SECTION – II

Multiple Correct Choice Type

This section contains 5 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

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5. For the reduction of NO_3^- ion in an aqueous solution, E^0 is +0.96 V. Values of E^0 for some metal ions are given below

$$V^{2+}(aq) + 2e^- \rightarrow V \qquad E^0 = -1.19\ V$$

$$Fe^{3+}(aq) + 3e^- \rightarrow Fe \qquad E^0 = -0.04\ V$$

$$Au^{3+}(aq) + 3e^- \rightarrow Au \qquad E^0 = +1.40\ V$$

$$Hg^{2+}(aq) + 2e^- \rightarrow Hg \qquad E^0 = +0.86\ V$$

The pair(s) of metals that is(are) oxidized by NO_3^- in aqueous solution is(are)

(A) V and Hg (B) Hg and Fe

(C) Fe and Au (D) Fe and V

Answer ● ● ○ ●

(A) (B) (C) (D)

6. Among the following, the state function(s) is(are)

(A) Internal energy (B) Irreversible expansion work

(C) Reversible expansion work (D) Molar enthalpy

Answer ● ○ ○ ●

(A) (B) (C) (D)

7. In the reaction

$$2\mathbf{X} + B_2H_6 \rightarrow [BH_2(\mathbf{X})_2]^+ [BH_4]^-$$

the amine(s) **X** is(are)

(A) NH_3 (B) CH_3NH_2

(C) $(CH_3)_2NH$ (D) $(CH_3)_3N$

Answer ● ● ● ○

(A) (B) (C) (D)

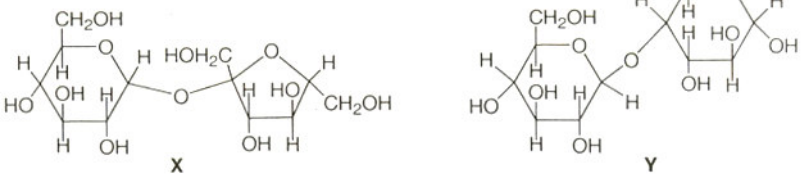
8. The nitrogen oxide(s) that contain(s) N-N bond(s) is(are)

(A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5

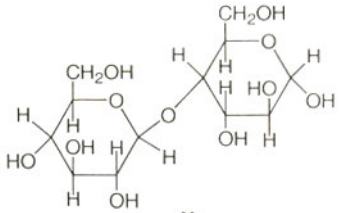
Answer ● ● ● ○

(A) (B) (C) (D)

9. The correct statement(s) about the following sugars **X** and **Y** is(are)



X



Y

(A) **X** is a reducing sugar and **Y** is a non-reducing sugar

(B) **X** is a non-reducing sugar and **Y** is a reducing sugar

(C) The glucosidic linkages in **X** and **Y** are α and β, respectively

(D) The glucosidic linkages in **X** and **Y** are β and α, respectively

Answer

○ ● ● ○

(A) (B) (C) (D)

SECTION – III

Matrix – Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following.

	p	q	r	s	t
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

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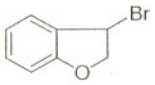
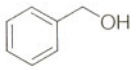
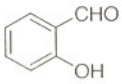
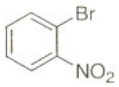
10. Match each of the reactions given in **Column I** with the corresponding product(s) given in **Column II**.

Column I	**Column II**
(A) $Cu + dil\ HNO_3$	(p) NO
(B) $Cu + conc\ HNO_3$	(q) NO_2
(C) $Zn + dil\ HNO_3$	(r) N_2O
(D) $Zn + conc\ HNO_3$	(s) $Cu(NO_3)_2$
	(t) $Zn(NO_3)_2$

Answer

	p	q	r	s	t
A	●	○	○	●	○
B	○	●	○	●	○
C	○	○	●	○	●
D	○	●	○	○	●

11. Match each of the compounds given in **Column I** with the reaction(s), that they can undergo, given in **Column II**.

Column I	Column II
(A) 	(p) Nucleophilic substitution
(B) 	(q) Elimination
(C) 	(r) Nucleophilic addition
(D) 	(s) Esterification with acetic anhydride
	(t) Dehydrogenation

Answer

	p	q	r	s	t
A	●	●	○	○	●
B	●	○	○	●	●
C	○	○	●	●	○
D	●	○	○	○	○

SECTION – IV

Integer Answer Type

This section contains 8 questions. The answer to each of the questions is a single–digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:

X	Y	Z	W
0	●	0	0
1	1	1	1
2	2	2	●
3	3	3	3
4	4	4	4
5	5	5	5
●	6	6	6
7	7	7	7
8	8	8	8
9	9	●	9

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12. In a constant volume calorimeter, 3.5 g of a gas with molecular weight 28 was burnt in excess oxygen at 298.0 K. The temperature of the calorimeter was found to increase from 298.0 K to 298.45 K due to the combustion process. Given that the heat capacity of the calorimeter is 2.5 kJ K^{-1}, the numerical value for the enthalpy of combustion of the gas in kJ mol^{-1} is

Answer 0 1 2 3 4 5 6 7 8 **9**

13. At 400 K, the root mean square (rms) speed of a gas **X** (molecular weight = 40) is equal to the most probable speed of gas **Y** at 60 K. The molecular weight of the gas **Y** is

Answer 0 1 2 3 **4** 5 6 7 8 9

14. The dissociation constant of a substituted benzoic acid at 25°C is 1.0×10^{-4}. The pH of a 0.01 M solution of its sodium salt is

Answer 0 1 2 3 4 5 6 7 **8** 9

15. The total number of α and β particles emitted in the nuclear reaction $^{238}_{92}U \rightarrow {}^{214}_{82}Pb$ is

Answer 0 1 2 3 4 5 6 7 **8** 9

16. The oxidation number of Mn in the product of alkaline oxidative fusion of MnO_2 is

Answer 0 1 2 3 4 5 **6** 7 8 9

17. The number of water molecule(s) directly bonded to the metal centre in $CuSO_4 \cdot 5H_2O$ is

Answer 0 1 2 3 **4** 5 6 7 8 9

18. The coordination number of Al in the crystalline state of $AlCl_3$ is

Answer 0 1 2 3 4 5 **6** 7 8 9

19. The total number of cyclic structural as well as stereo isomers possible for a compound with the molecular formula C_5H_{10} is

Answer 0 1 2 3 4 5 6 **7** 8 9

PART II : MATHEMATICS

SECTION – I

Single Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

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20. If the sum of first n terms of an A.P. is cn^2, then the sum of squares of these n terms is

(A) $\frac{n(4n^2-1)c^2}{6}$ (B) $\frac{n(4n^2+1)c^2}{3}$

(C) $\frac{n(4n^2-1)c^2}{3}$ (D) $\frac{n(4n^2+1)c^2}{6}$

Answer ○ ○ ● ○

(A) (B) (C) (D)